

ATOMIC STRUCTURE

1. The electronic configuration of an atom/ion can be defined by–
 - (1) Aufbau principal
 - (2) Hund's rule
 - (3) pauli exclusion principle
 - (4) All the above
2. Angular momentum of electrons in fourth orbit is–
 - (1) 0
 - (2) $\frac{2h}{\pi}$
 - (3) $\frac{3h}{2\pi}$
 - (4) $\left(\frac{3}{2}\right)^{\frac{1}{2}} \frac{h}{\pi}$
3. If Aufbau rule is not used, 19th electron in Sc ($Z = 21$) will have:–
 - (1) $n = 3, \ell = 0$
 - (2) $n = 3, \ell = 1$
 - (3) $n = 3, \ell = 2$
 - (4) $n = 4, \ell = 0$
4. The energy of a radiation of wavelength 8000 Å is E_1 and energy of a radiation of wavelength 16000 Å is E_2 . What is the relation between these two:–
 - (1) $E_1 = 6 E_2$
 - (2) $E_1 = 2 E_2$
 - (3) $E_1 = E_2$
 - (4) $E_1 = \frac{E_2}{2}$
5. Simultaneous determination of exact position and momentum of an electron in an atom is:–
 - (1) Possible
 - (2) Impossible
 - (3) Sometimes possible sometimes impossible
 - (4) None of these
6. When the value of $(n + \ell)$ is not more than 3 which of the following subshells is not possible to exist?
 - (1) 2s
 - (2) 3s
 - (3) 3p
 - (4) 2p
7. Given Rydberg constant $R = 10^5 \text{ cm}^{-1}$. Supposing if electron jumps from M shell to K shell of H-atom, the frequency of the radiation emitted in cycle/s would be–
 - (1) $\frac{8}{9} \times 10^5$
 - (2) $\frac{8}{3} \times 10^{15}$
 - (3) $\frac{8}{3} \times 10^{11}$
 - (4) $\frac{8}{9} \times 10^{15}$
8. Which of the following will violates Aufbau principle as well as Pauli's exclusion principle–
 - (1) $\begin{array}{|c|c|c|} \hline 1s & 2s & 2p \\ \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow \uparrow \square \\ \hline \end{array}$
 - (2) $\begin{array}{|c|c|c|} \hline 1s & 2s & 2p \\ \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow \uparrow\downarrow \uparrow \\ \hline \end{array}$
 - (3) $\begin{array}{|c|c|c|} \hline 1s & 2s & 2p \\ \hline \uparrow\downarrow & \uparrow & \uparrow\uparrow \uparrow \\ \hline \end{array}$
 - (4) None of these
9. Energy of electron of hydrogen atom in second Bohr orbit is–
 - (1) $-5.44 \times 10^{-19} \text{ J}$
 - (2) $-5.44 \times 10^{-19} \text{ KJ}$
 - (3) $-5.44 \times 10^{-19} \text{ cal}$
 - (4) $-5.44 \times 10^{-19} \text{ ev}$
10. In H-atom if electron falls from $n = 3$ to $n = 2$, then emitted energy is–
 - (1) 10.2 eV
 - (2) 12.09 eV
 - (3) 1.9 eV
 - (4) 0.65 eV
11. de-Broglie equation is–
 - (1) $h = \frac{\lambda}{mv}$
 - (2) $E = hv$
 - (3) $E = mc^2$
 - (4) $\lambda = \frac{h}{mv}$
12. The shape of an orbital is given by the quantum number–
 - (1) n
 - (2) ℓ
 - (3) m
 - (4) s
13. An ion has 18 electrons in the outer most shell, it is:
 - (1) Cu^+
 - (2) Th^{4+}
 - (3) Cs^+
 - (4) K^+
14. The energy corresponding to second line of Balmer series for hydrogen atom will be–
 - (1) 12.1 eV
 - (2) 1.89 eV
 - (3) 2.55 eV
 - (4) 13.6 eV
15. If ionization potential of hydrogen like species is 36 V, then the second excitation potential of electron will be–
 - (1) 10.2 V
 - (2) 12 V
 - (3) 27 V
 - (4) 32 V
16. If the value of principal quantum number is 4, the possible values for magnetic quantum number will be–
 - (1) 4
 - (2) 16
 - (3) 32
 - (4) 9

17. Bohr's model could explain successfully:-
 (1) The spectrum of helium
 (2) The spectrum of species containing only one electron
 (3) The spectrum of multi-electron
 (4) The spectrum of hydrogen molecule
18. Calculate de Broglie wavelength of a proton (mass, $m = 1.6 \times 10^{-27}$ kg) moving with kinetic energy of 0.04 eV-
 (1) 146 Å (2) 14.6 Å
 (3) 1460 Å (4) 1.46 Å
19. Consider the following statements-
 (A) Electron density in XY plane in $3d_{x^2-y^2}$ is zero
 (B) Electron density in XY plane in $3d_{z^2}$ orbital is zero
 (C) 2s orbital has only one spherical node
 (D) For $2P_z$ orbital YZ is the nodal plane
 The correct statements are-
 (1) (b) and (c)
 (2) (a), (b), (c), (d)
 (3) only (b)
 (4) only (c)
20. If the electron can be located in the orbit with in 0.1 Å. What is the uncertainty in its velocity-
 (1) $5.76 \times 10^7 \text{ ms}^{-1}$
 (2) $0.576 \times 10^7 \text{ ms}^{-1}$
 (3) $0.576 \times 10^{-4} \text{ ms}^{-1}$
 (4) $5.76 \times 10^{-4} \text{ ms}^{-1}$
21. Which is not the characteristic property of plank's quantum theory?
 (1) Energy is neither absorbed nor emitted in simple multiples of quanta
 (2) Radiation is related to energy
 (3) Energy is not radiated continuously but in small packages of energy called quantum
 (4) The energy of a quantum is proportional to its frequency
22. Consider the ground state of Cr atom ($z = 24$). The number of electrons with azimuthal quantum numbers, $\ell = 1$ and 2 are respectively:-
 (1) 16 and 4 (2) 12 and 5
 (3) 12 and 4 (4) 16 and 5
23. The energy of electron in 3rd orbit of hydrogen atom is-
 (1) - 1311.8 KJ/mol (2) -82 KJ/mol
 (3) -145.7 KJ/mol (4) -327.9 KJ/mol
24. A certain radiostation broadcasts at a frequency of 900 KHz. The wavelength of electromegnetic radiations broadcast by radiostation is-
 (1) 900 m (2) 270 m
 (3) 2.70 km (4) 330 m
25. The radius of Bohr's first excited state is a_0 . The electron in n^{th} orbit has a radius:
 (1) $\frac{n^2 a_0}{4}$ (2) $\frac{a_0}{n}$
 (3) $n^2 a_0$ (4) $\frac{a_0}{n^2}$